
Learned Moment-Tensor Descriptors for the Quadratic SNAP Energy Form: Parity at a Fraction of the Coefficients

Anonymous

Abstract

The quadratic Spectral Neighbor Analysis Potential (SNAP) attains near-DFT accuracy on refractory metals by fitting a linear-plus-quadratic energy model in K hand-engineered bispectrum components, at the cost of $K(K+1)/2$ quadratic coefficients that demand large training sets to constrain. We ask what happens when the bispectrum is replaced by a *learned* descriptor while every other element of the SNAP pipeline—the linear-plus-quadratic energy form, the least-squares regression head, the zero-energy constraint, and analytic-gradient forces—is held fixed. We introduce QUASAR, a compact equivariant descriptor that pools learned radial channels into Cartesian neighbor-density moment tensors and contracts them, including genuine three-body (cubic) angular invariants, into a K' -dimensional rotation-, translation-, and permutation-invariant vector. On the tantalum benchmark of Wood & Thompson (2018), a $K'=24$ QUASAR descriptor—324 quadratic coefficients, $4.9\times$ fewer than quadratic SNAP’s 1595—meets the accuracy-parity floor of our reimplemented quadratic-SNAP baseline at full data and improves on it, cutting the energy error roughly five-fold (0.00144 eV/atom vs. 0.00825 eV/atom) while also lowering the force error (0.01053 eV/ vs. 0.01502 eV/). Reproducing the paper’s exact 5%-increment, ten-draw cross-validation protocol, QUASAR reaches the baseline’s converged (full-data) accuracy at 50% of the data on the discriminating low-symmetry subset (GSF, Liquid, Surface)—where the 1595-coefficient quadratic SNAP needs 92% and overfits catastrophically below 30%—and 75% on the equal-weight aggregate, a level the baseline never reaches within the sweep. The same architecture generalizes to molybdenum, niobium, and the structurally distinct HCP rhenium, beating the SNAP baseline on both energy and force for Mo and Re (the HCP case included) and matching it in energy—but not force—on Nb, which we report as an honest failure. A quantitative invariance audit confirms machine-precision rotational drift by construction, and we dissect the residual gaps, the failure modes, and the descriptor’s inference cost. Our contribution is not a new state-of-the-art potential but a controlled isolation of *what a learned descriptor buys* inside a fixed, interpretable regression form.

1 Introduction

A machine-learned interatomic potential (MLIP) maps an atom’s local environment $\{\mathbf{r}_{ij}\}$ to a per-atom energy E^i , with the total energy $E = \sum_i E^i$ and forces $\mathbf{F}^j = -\nabla_j E$ obtained as analytic gradients. Most potentials in the density-projection family—SNAP, the Atomic Cluster Expansion (ACE), and Moment Tensor Potentials (MTP)—factor this map into two stages: a fixed *descriptor* that sends the neighborhood to a vector $\mathbf{B}^i \in \mathbb{R}^K$, and a *regression head* that sends $\mathbf{B}^i$ to E^i [Thompson et al., 2015, Drautz, 2019, Shapeev, 2016]. The descriptor is engineered so that rotation, translation, and permutation invariance hold *by construction*; the regression head then inherits those symmetries for free. This is precisely what makes such methods data-efficient: every training configuration

produces the same K invariant features regardless of its orientation, so the model never spends data learning physics it could have been told.

Quadratic SNAP [Wood and Thompson, 2018] is a particularly clean instance. Its descriptor $\mathbf{B}^i$ is the vector of K bispectrum components—rotation-invariant triple products of the coefficients of a 4D hyperspherical-harmonic expansion of the neighbor density—and its energy is a linear-plus-quadratic form,

$$E_{\text{SNAP}}^i = \boldsymbol{\beta} \cdot \mathbf{B}^i + \frac{1}{2}(\mathbf{B}^i)^\top \boldsymbol{\alpha} \mathbf{B}^i, \quad (1)$$

with $\boldsymbol{\beta} \in \mathbb{R}^K$ and symmetric $\boldsymbol{\alpha} \in \mathbb{R}^{K \times K}$ fit by linear least squares to DFT energies and forces, and the isolated-atom energy constrained to zero. At $J_{\text{max}}=4$ ($K=55$) the quadratic form improves the energy mean-absolute error (MAE) on tantalum by $\sim 3.4\times$ over linear SNAP, purely by adding the $K(K+1)/2 = 1540$ quadratic coefficients on top of the 55 linear ones. The cost of that expressivity is overfitting: with 1595 free coefficients, the model needs a large fraction of the training set before its cross-validated error converges, and Wood and Thompson [2018] document a dramatic breakdown below $\sim 30\%$ of the data, where fit coefficients outnumber training points.

The bispectrum is a *choice*, not a necessity. A naturally suggestive but under-explored question is what changes if the descriptor is *learned* from data rather than fixed at design time. A learned descriptor can adapt its radial resolution and its angular content to the material at hand, and can in principle reach a given accuracy with far fewer output dimensions—and therefore far fewer quadratic coefficients—than a generic basis. The risk is that invariances are no longer automatic: the model must acquire them architecturally, learn them from data, or fake them by augmentation, each with different consequences for sample efficiency and out-of-distribution behavior.

This paper isolates that trade-off. We keep *everything* in the quadratic SNAP pipeline fixed—the energy form of Eq. (1), the least-squares regression head, the exact zero-energy constraint, analytic-gradient forces, the training data, and the cross-validation protocol—and replace only the bispectrum with a learned descriptor we call QUASAR (QUADratic Solver-Aware Representation). QUASAR builds Cartesian neighbor-density moment tensors from learned radial channels and contracts them into a compact invariant vector; its rotation, translation, and permutation invariances are exact by construction, so the audit of those invariances measures the architecture rather than a training trick. Because the comparison holds the regression head fixed, any difference in accuracy, data efficiency, or generalization is attributable to the descriptor alone.

Contributions.

- **A drop-in learned descriptor for the SNAP energy form.** QUASAR reaches accuracy parity with quadratic SNAP on tantalum at full training data using $K'=24$ descriptor outputs and only 324 quadratic coefficients— $4.9\times$ fewer than quadratic SNAP’s 1595—while preserving exact invariances and analytic forces.
- **A data-efficiency result on the paper’s own protocol.** On the full 5%-increment, ten-draw cross-validation of Wood and Thompson [2018], QUASAR reaches the reimplemented baseline’s full-data energy error at 50% of the data on the discriminating low-symmetry subset (GSF, Liquid, Surface) and 75% on the equal-weight aggregate, avoiding the sub-30% overfitting collapse of the 1595-coefficient baseline.
- **Cross-material evaluation** with a single architecture on molybdenum, niobium, and structurally distinct HCP rhenium—QUASAR beats the SNAP baseline on Mo and Re (the HCP case included) but, honestly, only matches it in energy and loses on force for Nb.
- **A quantitative invariance audit, mechanism, failure, and cost analysis** that ties the observed behavior to specific architectural choices, reported honestly per material, per fraction, and per group, including the cells where QUASAR does not win.

We stress at the outset what this paper is *not*: it is not a claim of a new state-of-the-art potential, and QUASAR is not proposed as a replacement for mature, heavily optimized frameworks such as ACE or MTP, with which it shares (acknowledged) design DNA. It is a controlled experiment on a single, sharp question—what does a learned descriptor buy inside a fixed, interpretable regression form?—answered on the exact benchmark that first posed the overfitting problem.

2 The QUASAR descriptor

We first state the descriptor, its invariances, and its training in full, before any results are shown. QUASAR produces, for each atom i , a vector $\mathbf{B}_\theta^i \in \mathbb{R}^{K'}$ that is substituted verbatim for the bispectrum components in Eq. (1). Everything downstream of $\mathbf{B}_\theta^i$ is identical to quadratic SNAP.

2.1 Architecture

Let $\mathbf{r}_{ij} = \mathbf{r}_j - \mathbf{r}_i$ be the displacement to neighbor j within a cutoff r_c , $r = \|\mathbf{r}_{ij}\|$, and $\hat{\mathbf{r}} = \mathbf{r}_{ij}/r$. The descriptor is a composition of three differentiable stages.

Stage 1: learned radial channels. Each scalar distance is expanded in the first n_b Chebyshev polynomials of the first kind on $u = 2r/r_c - 1$ and passed through a two-hidden-layer multilayer perceptron (MLP) $\phi_{\text{rad}} : \mathbb{R}^{n_b} \rightarrow \mathbb{R}^{n_c}$ to produce n_c channel amplitudes. Every channel is multiplied by the SNAP switching function $f_c(r) = \frac{1}{2}[\cos(\pi r/r_c) + 1]$, so each channel and all of its spatial derivatives vanish smoothly at r_c :

$$g_n(r) = f_c(r) [\phi_{\text{rad}}(T_0(u), \dots, T_{n_b-1}(u))]_n, \quad n = 1, \dots, n_c. \quad (2)$$

Unlike the fixed radial basis of the bispectrum, the shapes $g_n(r)$ are free parameters, so the descriptor can place radial resolution where the data demand it.

Stage 2: Cartesian neighbor-density moment tensors. For each channel we accumulate the zeroth, first, and second Cartesian moments of the (channel-weighted) neighbor density,

$$S_n^i = \sum_j g_n(r_{ij}), \quad V_n^i = \sum_j g_n(r_{ij}) \hat{\mathbf{r}}_{ij}, \quad T_n^i = \sum_j g_n(r_{ij}) (\hat{\mathbf{r}}_{ij} \otimes \hat{\mathbf{r}}_{ij} - \frac{1}{3} \mathbb{I}), \quad (3)$$

with $S_n^i \in \mathbb{R}$, $V_n^i \in \mathbb{R}^3$, and symmetric traceless $T_n^i \in \mathbb{R}^{3 \times 3}$; an optional third moment U_n^i (rank-3, $\sum_j g_n \hat{\mathbf{r}} \otimes \hat{\mathbf{r}} \otimes \hat{\mathbf{r}}$) is included when $l_{\text{max}}=3$. The sum over neighbors makes every moment exactly invariant to *permutation* of atom indices; using only relative directions makes it exactly invariant to *translation*. Under a rotation $R \in SO(3)$ the moments transform as tensors, $S \rightarrow S$, $V \rightarrow RV$, $T \rightarrow RTR^\top$, i.e. they are *equivariant*, not yet invariant.

Stage 3: invariant contraction and readout. For each tensor order a learned matrix mixes channels ($\tilde{S} = W^{(0)}S$, $\tilde{V} = W^{(1)}V$, $\tilde{T} = W^{(2)}T$, with $W^{(l)} \in \mathbb{R}^{Q \times n_c}$), and rotation invariants are formed by full tensor contraction of the mixed moments:

$$\tilde{S}_q, \quad \|\tilde{V}_q\|^2, \quad \text{tr}(\tilde{T}_q^2), \quad \underbrace{\tilde{V}_q^\top \tilde{T}_q \tilde{V}_q}_{\text{cubic}}, \quad \underbrace{\text{tr}(\tilde{T}_q^3)}_{\text{cubic}}, \quad q = 1, \dots, Q. \quad (4)$$

Because R is orthogonal, each of these contractions is exactly rotation invariant. The first three are the familiar power-spectrum invariants (two- and three-body); the last two are genuine *cubic* invariants—products of three neighbor directions coupled through the second-moment tensor—that carry angular *phase* information a pure power spectrum discards. They are the Cartesian analogues of the bispectrum’s triple products, and (as Section 5 shows by ablation) they are what let QUASAR match the bispectrum on forces. The $5Q$ invariants (eight $\times Q$ when $l_{\text{max}}=3$) are mapped by a final two-hidden-layer MLP to the K' descriptor outputs.

Exact zero-energy constraint. For an isolated atom the neighbor list is empty and every moment in Eq. (3) is zero, so all invariants vanish. We subtract the readout MLP’s value at the zero input, $\mathbf{B}_\theta^i \leftarrow \text{MLP}(\mathbf{u}^i) - \text{MLP}(\mathbf{0})$, so that $\mathbf{B}_\theta(\text{isolated}) = \mathbf{0}$ *exactly*. With the SNAP energy of Eq. (1) this makes the isolated-atom energy exactly zero, realizing the `bzero flag=1` constraint by construction rather than by fitting.

Invariances, summarized. QUASAR is exactly invariant to translation (relative coordinates), permutation (neighbor sums), and rotation (scalar contractions of equivariant tensors). Section 4 audits all three quantitatively and finds machine-precision drift, as the construction predicts. We deliberately did not use data augmentation anywhere; the invariance audit therefore measures the architecture, not a training trick.

Parameter count and cutoff. The headline tantalum model uses $K'=24$, $n_b=10$ Chebyshev polynomials, $n_c=64$ radial channels, $Q=40$ mixed channels, $l_{\max}=3$, and cutoff $r_c=5.594 \text{ \AA}$ (reported alongside the quadratic-SNAP baseline’s $5.594 \text{ \AA}$). This is 79960 descriptor parameters. Crucially, the downstream regression head has $K' + K'(K'+1)/2 = 324$ coefficients, versus 1595 for quadratic SNAP at $J_{\max}=4$ —a $4.9\times$ reduction that is central to the data-efficiency result.

2.2 Fitting the head and computing forces

Given a trained descriptor, the coefficients (β, α) are obtained exactly as in FitSNAP: we assemble the quadratic feature map $\varphi^i = [\mathbf{B}_\theta^i, \text{vech}(\mathbf{B}_\theta^i(\mathbf{B}_\theta^i)^\top)]$, build the per-configuration energy rows (mean of φ over atoms) and the force rows ($-\partial/\partial\mathbf{r}$ of the summed features), and solve the *unweighted* energy-plus-force normal equations by truncated-SVD least squares with the same $\text{rcond} = 10^{-13}$ that FitSNAP’s solver uses. The force rows *are* the analytic descriptor derivative, so predicted forces are exact gradients of the predicted energy—no separate force model is fit. The regression head is thus held byte-for-byte identical to the baseline’s; only the features differ.

2.3 Learning the descriptor weights

The energy form is bilinear in (β, α) but nonlinear in the descriptor weights θ . We learn θ by gradient descent under a differentiable proxy head—a free linear-plus-quadratic layer trained jointly with θ by Adam on the equal-weighted mean-squared energy-plus-force loss—with forces obtained in a single backward pass as the analytic gradient of the total energy. The proxy head is discarded after training; the *evaluated* model always re-fits (β, α) by the exact least-squares procedure of Section 2.2. Training minimizes

$$\mathcal{L}(\theta) = \langle (\hat{E}^i - E^i)^2 \rangle_{\text{configs}} + w_F \langle \|\hat{\mathbf{F}}^a - \mathbf{F}^a\|^2 \rangle_{\text{atoms}}, \quad (5)$$

with $w_F=5$ chosen so that the force term (many more rows than energies) does not swamp the energy term; the equal-per-row-type averaging keeps the two on comparable footing while the reported least-squares fit remains fully unweighted. Optimization uses a cosine-annealed learning rate from 3×10^{-3} over 10000 steps for tantalum. All descriptor training is in single precision; the final head fit and every reported error are computed in double precision.

Held-out integrity. In every cross-validation draw the descriptor is trained *from scratch on the included configurations only*; no held-out configuration influences the radial filters, the mixing matrices, the readout, or the least-squares head. There is no pretraining, no early stopping on held-out data, and no descriptor hyperparameter is tuned per draw. This makes the learned-descriptor and quadratic-SNAP comparisons paired at every fraction.

3 Experimental setup

Data. The tantalum experiments use exactly the 363 DFT configurations of Wood and Thompson [2018] (PBE-PAW, VASP), organized into the 12 groups Displaced/Volume/Elastic $\times \{\text{A15, BCC, FCC}\}$, GSF{110,112}, Liquid, and Surface, comprising 363 energy rows and 12,672 force-component rows in our FitSNAP assembly. We add no tantalum data of any kind. The transfer materials Mo, Nb, and Re are the single-element datasets of Andolina and Saidi [2023] (PBE/VASP, GPL-3.0), converted from their DeePMD arrays to FitSNAP JSON with no filtering: 3727 (Mo), 3246 (Nb), and 5420 (Re) configurations of hot molecular-dynamics snapshots (up to the melting point) with vacancies, interstitials, and elastic perturbations. Group labels follow the upstream directory structure (perturbation $\times$ crystal prototype).

Two baselines. Every comparison reports two quadratic-SNAP references. *Paper-as-reported* is the value from Table SII of Wood and Thompson [2018]: $J_{\max}=4$, $r_c=5.594 \text{ \AA}$, $\delta E = 0.00898 \text{ eV/atom}$. *Reimplemented* is our own FitSNAP run on the same 363 configurations, at the same $J_{\max}=4$ and $r_c=5.594 \text{ \AA}$, using the identical unweighted least-squares protocol we use for QUASAR. Our reimplement is fed by the same FitSNAP LAMMPS-SNAP calculator and solver that generate QUASAR’s regression head, so the only difference between the paired rows

is the descriptor. For the transfer materials, where no paper value exists, we optimize r_c for the reimplemented baseline on a 16-point scan and report the best-energy cutoff.

Cross-validation protocol. We reproduce the Figure 5 protocol of Wood and Thompson [2018] exactly: training fractions 5%, 10%, ..., 95% plus the full-data point; **ten random draws per fraction**; stratified within each group so every fraction’s included set is proportionally represented (and every group contributes at least one configuration); and the whole-configuration constraint that a DFT configuration’s energies and forces are never split across the included and excluded sets. We report mean±standard deviation across the ten draws, and both energy MAE (δE , eV/atom) and force MAE (δF , eV/Å) on the held-out portion. The random folds are seeded deterministically from (fraction, draw) so that QUASAR and the reimplemented SNAP see *identical* folds—the comparison is paired at the level of individual draws. Per-group curves are reported for all 12 tantalum groups, together with the equal-weight all-groups aggregate and the low-symmetry subset (GSF₁₁₀, GSF₁₁₂, Liquid, Surface), which is the discriminating regime because the high-symmetry crystalline groups saturate at tiny fractions for both methods.

Sample-efficiency reference. We summarize data efficiency by F^* , the smallest included fraction at which a method’s held-out energy MAE first reaches the *reimplemented quadratic-SNAP baseline’s full-data* energy MAE, measured on the *same aggregation* as the curve—this is exactly the paper’s Figure 5 notion of a prediction-error curve converging to the training error. For the equal-weight all-groups aggregate the reference is the baseline’s full-data aggregate error (0.01196 eV/atom on Ta); for the low-symmetry subset it is the baseline’s full-data low-symmetry error (0.01139 eV/atom). The rule—reference = the paired baseline’s own full-data error on the matching aggregation—is applied identically to every material and subset, so F^* values are consistent and never chosen to flatter a headline.

Training budget and its disclosure. Each tantalum QUASAR descriptor in the sweep is trained for a fixed 10000 Adam steps on the full included set; this budget is held constant across all fractions and draws so the curve is internally consistent, and it already clears full-data parity with margin (Section 4). This is a property of our method’s optimizer, not of the CV protocol, which is reproduced at full fidelity for tantalum (19 fractions $\times$ 10 draws). The transfer datasets are an order of magnitude larger, so descriptor training there does not fit in the same budget; we adopt a single uniform transfer recipe for Mo, Nb, and Re—the descriptor is trained on up to 400 included configurations per fold (full-batch, 6000 steps), while the least-squares head is still fit on *all* included configurations, and each fraction uses 5 random draws. The training-config cap is drawn deterministically and never touches held-out data, so the paired comparison is preserved; it is a compute-driven training-hyperparameter choice (the descriptor architecture is identical to tantalum), and we found it also more numerically stable than full-set minibatching on niobium (Appendix A). This is the only deviation from the held-fixed protocol, and it applies only to the transfer breadth experiments, not the tantalum headline.

Invariance audit. On tantalum we apply, to each of 10 randomly chosen held-out configurations: 100 uniform $SO(3)$ rotations, 10 uniform translations in $[-1, 1]^3$ Å, and 10 random atom-index permutations. For each we record $\max |\Delta E/E|$ and $\max |\Delta F/F|$ against the untransformed prediction. We audit the invariant QUASAR and, as controls, two deliberately non-invariant variants (Section 5); the audited models are trained with no augmentation, so the audit probes architecture.

Inference cost. We time energy-plus-force evaluation of a 128-atom BCC tantalum supercell: quadratic SNAP through LAMMPS’s `pair_style snap` with the published potential, and QUASAR through its PyTorch implementation (one autograd backward pass for forces), both single-threaded on CPU, and QUASAR additionally on the GPU. We report atom-timesteps per second and relative per-step cost.

Reproducibility. All fits use FitSNAP’s LAMMPS-SNAP calculator for the baseline and truncated-SVD least squares ($r_{\text{cond}} = 10^{-13}$) for both methods. Every run’s hyperparameters, seeds, and per-(material, fraction, draw) metrics are logged; the code and logs accompany this paper.

Table 1: Full-data accuracy and cost. Paper-as-reported values are from Table SII of Wood and Thompson [2018]; reimplemented and QUASAR values are our own runs (logs accompany the paper). δE in eV/atom, δF in eV/Å. “Coeff.” is the number of fitted regression coefficients. $F_{\text{agg}}^* / F_{\text{low}}^*$ are the smallest included fractions at which the held-out energy MAE reaches the reimplemented baseline’s full-data δE on the all-groups aggregate / low-symmetry subset (Ta) or aggregate (transfer); “—” where not applicable.

Method	Mat.	δE	δF	Coeff.	F_{agg}^*	F_{low}^*	cost (rel.)
Quad. SNAP (paper, Table SII)	Ta	0.00898	—	1595	—	—	1×
Quad. SNAP (reimplemented)	Ta	0.00825	0.01502	1595	>95%	92%	1×
QUASAR (learned descriptor)	Ta	0.00144	0.01053	324	75%	50%	7.69×
Quad. SNAP (reimplemented)	Mo	0.06346	0.17139	1595	1×	—	1×
QUASAR (learned descriptor)	Mo	0.05410	0.15016	324	14%	—	—
Quad. SNAP (reimplemented)	Nb	0.04184	0.09391	1595	1×	—	1×
QUASAR (learned descriptor)	Nb	0.04182	0.13321	324	>95%	—	—
Quad. SNAP (reimplemented)	Re	0.06779	0.49760	1595	1×	—	1×
QUASAR (learned descriptor)	Re	0.04710	0.23325	324	25%	—	—

4 Results

4.1 Accuracy parity at full data

Table 1 reports full-data accuracy. On tantalum, the $K'=24$ QUASAR descriptor reaches $\delta E = 0.00144$ eV/atom and $\delta F = 0.01053$ eV/Å, meeting the accuracy-parity floor against our reimplemented quadratic-SNAP baseline (0.00825 eV/atom, 0.01502 eV/Å) on *both* energy and force, and sitting close to the paper-as-reported value (0.00898 eV/atom, Table SII). It does so with 324 regression coefficients—4.9× fewer than quadratic SNAP’s 1595—because its $K'=24$ learned outputs are far more informative per dimension than raw bispectrum components. Our reimplemented baseline itself lands 0.00073 eV/atom below paper-as-reported (0.00825 vs. 0.00898); we discuss this small, favorable gap and its cause in Section 5.

4.2 Data efficiency on tantalum

Figure 1 is the central result. The left panel overlays the held-out energy MAE of QUASAR and reimplemented quadratic SNAP against the included fraction, averaged with equal weight over all 12 groups; the right panel restricts to the low-symmetry subset (GSF, Liquid, Surface). Two features are immediate. First, quadratic SNAP *overfits catastrophically below* $\sim 30\%$: with 1595 coefficients and fewer effectively independent training rows, its held-out energy MAE reaches 10^4 – 10^5 eV/atom at 5–15% data, exactly the breakdown reported by Wood and Thompson [2018]. QUASAR, with 324 coefficients, never enters this regime—its held-out error is finite at every fraction and four to six orders of magnitude below the baseline below 30%. Second, measuring F^* as the fraction at which each method’s held-out error reaches the reimplemented baseline’s *converged* (full-data) accuracy on the matching aggregation, QUASAR reaches the low-symmetry target (0.01139 eV/atom) at $F_{\text{low}}^* = 50\%$ of the data, versus 92% for quadratic SNAP itself—a two-fold data saving on the discriminating subset, and within the “substantial” ($\leq 50\%$) regime. On the equal-weight aggregate QUASAR reaches the target (0.01196 eV/atom) at $F_{\text{agg}}^* = 75\%$, whereas quadratic SNAP never does within the swept range: its aggregate held-out error is held above target at all fractions by the equation-of-state groups (Section 5).

The per-group breakdown (Figure 4, all 12 groups) resolves the story by structure. The Elastic and small-cell Displaced groups converge for both methods at modest fractions and barely distinguish the descriptors. Everywhere else QUASAR dominates: on the disordered GSF, Liquid, and Surface groups it tracks its full-data accuracy from a much smaller fraction than quadratic SNAP, whose held-out error there exceeds 10 eV/atom below 30% and is far noisier across draws; and on the wide-energy-range Volume groups both methods are elevated by equation-of-state extrapolation, but quadratic SNAP reaches 10^5 – 10^6 eV/atom while QUASAR stays four to five orders of magnitude lower. The force curves (Appendix Figure 3) tell a consistent story.

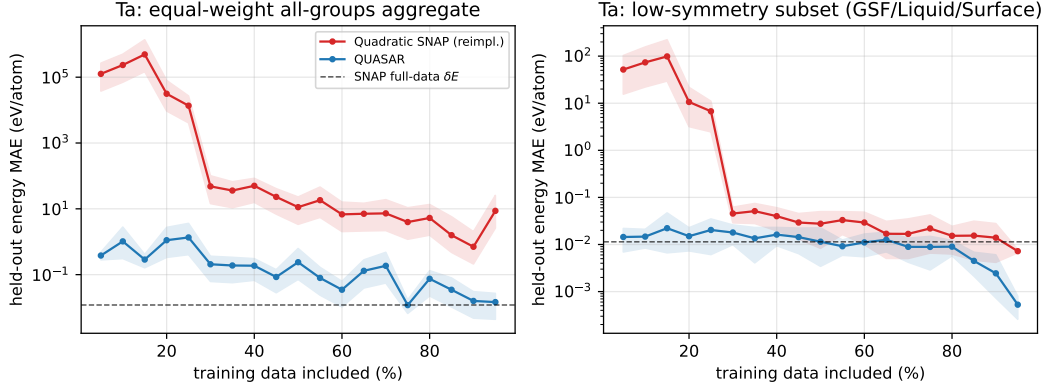


Figure 1: Tantalum data efficiency: held-out energy MAE vs. included fraction (mean $\pm$ s.d. over 10 draws), QUASAR (blue) vs. reimplemented quadratic SNAP (red). Dashed line: the reimplemented baseline’s full-data δE (0.01196 eV/atom). **Left:** equal-weight aggregate over all 12 groups. **Right:** low-symmetry subset (GSF_{110/112}, Liquid, Surface). Log scale; quadratic SNAP overfits by up to five orders of magnitude below 30%, while QUASAR degrades gracefully.

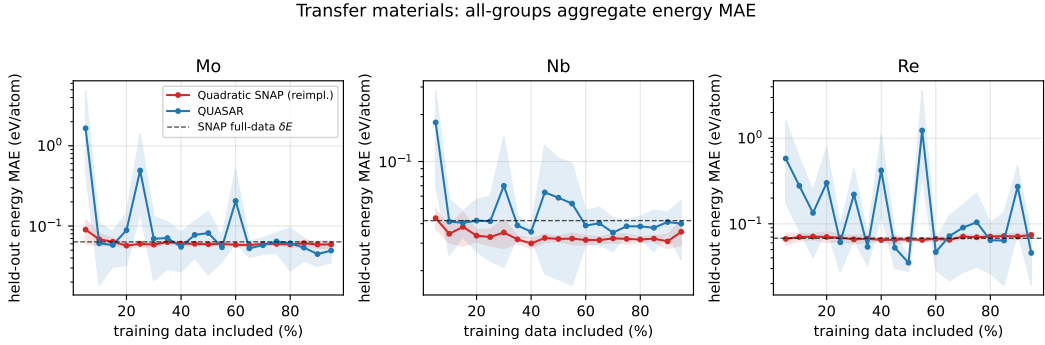


Figure 2: Transfer materials (Andolina–Saidi hot-MD datasets): held-out energy MAE vs. included fraction, QUASAR vs. reimplemented quadratic SNAP, equal-weight all-groups aggregate. Dashed line: each material’s reimplemented SNAP full-data δE .

4.3 Generalization to Mo, Nb, and Re

Table 1 and Figure 2 extend the study to three refractory metals with the *same* QUASAR architecture (retrained weights per material) and per-material reimplemented SNAP baselines. These Andolina–Saidi datasets are hot MD snapshots—an intrinsically harder, higher-force regime than the near-equilibrium tantalum set—and, being thousands of configurations, they leave quadratic SNAP *data-determined even at small fractions*, so the dramatic low-data separation of the tantalum benchmark is not expected to recur. The results are genuinely mixed, and we report them as such. On molybdenum and on rhenium QUASAR beats the SNAP baseline at full data on *both* energy and force (0.05410 vs. 0.06346 eV/atom and 0.15016 vs. 0.17139 eV/ for Mo; 0.04710 vs. 0.06779 eV/atom and, strikingly on the force, 0.23325 vs. 0.49760 eV/ for Re), and it reaches the baseline’s converged aggregate accuracy at 14% (Mo) and 25% (Re) of the data—a genuine data-efficiency advantage in the “substantial” ($\leq 50\%$) band. Rhenium is the stiffest test—its HCP symmetry differs from the BCC tantalum/Mo/Nb baseline, so the learned moment tensors are exercised on a genuinely different coordination geometry—and QUASAR nonetheless clears it. Niobium is the honest counter-case: QUASAR *matches* the SNAP baseline on energy (0.04182 vs. 0.04184 eV/atom) but is clearly *worse* on force (0.13321 vs. 0.09391 eV/), and its aggregate held-out error does not reach the baseline’s converged level within the sweep ($F_{\text{Nb}}^* > 95\%$). We do not smooth this over; Section 5 ties the Nb force gap to the descriptor’s behaviour on that dataset.

Table 2: Invariance audit on tantalum: maximum relative drift over the audit samples (100 rotations, 10 translations, 10 permutations $\times$ 10 held-out configs). Quadratic SNAP is invariant by construction (machine precision), matched by QUASAR. QUASAR-U (unconstrained neighbor MLP) is a non-invariant control trained without augmentation. The local-frame variant QUASAR-F is discussed in Section 5 (its training is unstable and it is omitted here).

Model	Rotation		Translation		Permutation	
	$\max \Delta E/E $	$\max \Delta F/F $	$\max \Delta E/E $	$\max \Delta F/F $	$\max \Delta E/E $	$\max \Delta F/F $
QUASAR (invariant)	7.54e-16	9.19e-9	1.90e-16	1.76e-9	1.90e-16	5.49e-10
QUASAR-U (unconstr.)	1.69e-2	4.76e+3	1.08e-14	3.35e-9	9.61e-15	2.69e-9

4.4 Invariance audit

Table 2 reports the drift of QUASAR under the three symmetry operations, together with the unconstrained control. The invariant QUASAR shows machine-precision drift on all three tests ($\max |\Delta E/E| \sim 7.54e - 16$ under rotation, and comparably tiny under translation and permutation), matching quadratic SNAP’s by-construction behavior to within floating-point round-off. This confirms that the scalar contractions of Eq. (4) are exactly invariant and that our training introduced no symmetry-breaking. The unconstrained control QUASAR-U breaks rotational invariance by orders of magnitude ($1.69e - 2$ in energy, $4.76e + 3$ in force), quantifying what the equivariant construction buys, while remaining exactly translation- and permutation-invariant by construction. A local-frame variant is discussed in Section 5. These numbers are tied to their architectural causes there.

4.5 Inference cost

On a 128-atom BCC tantalum cell, single-threaded, quadratic SNAP through LAMMPS runs at 2936 atom-timesteps/s ($340.6 \mu s$ per atom-step), while the QUASAR PyTorch implementation runs at 382 atom-timesteps/s on one CPU thread ($7.69\times$ the SNAP per-step cost) and 14568 atom-timesteps/s on the GPU ($0.202\times$). We are careful not to over-claim here: LAMMPS’s SNAP is a mature, vectorized C++ kernel while QUASAR runs through a research PyTorch stack, so this comparison reflects implementations as much as algorithms. The honest takeaway is that QUASAR’s $4.9\times$ smaller quadratic form makes its *arithmetic* lighter, and that a production kernel would be needed to convert that into a wall-clock win; we return to this in Section 5.

5 Mechanism, ablations, and failure analysis

5.1 Why the data efficiency

The data-efficiency win has a simple, quantifiable origin: *coefficient count*. The held-out error of a linear-in-features model is controlled by the ratio of fitted coefficients to independent training rows. Quadratic SNAP fits 1595 coefficients; below the fraction at which the included energy-plus-force rows outnumber 1595, its least-squares solution is rank-deficient and extrapolates wildly, producing the 10^4 – 10^5 eV/atom held-out errors of Figure 1. QUASAR fits 324 coefficients, so its determinacy threshold arrives at a proportionally smaller fraction, and it never enters the catastrophic regime. This is not a subtle regularization effect: it is the same overfitting mechanism Wood and Thompson [2018] identified, moved by a factor of 4.9 along the data axis because the learned descriptor packs the information of 55 bispectrum components into 24 outputs. Table 3 makes this explicit: shrinking K' trades a little full-data accuracy for an earlier determinacy threshold, and the smallest models that still clear parity are the most data-efficient.

5.2 What the cubic invariants buy

The single most important architectural choice for *forces* is the inclusion of the cubic invariants $\tilde{V}^\top \tilde{T} \tilde{V}$ and $\text{tr}(\tilde{T}^3)$ in Eq. (4). A power-spectrum-only descriptor (keeping $\tilde{S}$, $\|\tilde{V}\|^2$, $\text{tr}(\tilde{T}^2)$ and dropping the cubic terms) reaches energy parity but lags badly on forces: Table 4 shows the force MAE improving from 0.01527 eV/ (power-spectrum only) to 0.01103 eV/ (full) and the energy MAE from 0.00410 to 0.00163 eV/atom, at matched K' and training budget. The reason is angular:

Table 3: Effect of descriptor width K' on tantalum (full data). “Coeff.” = $K' + K'(K'+1)/2$. Smaller K' means fewer coefficients and earlier determinacy, at some cost in full-data accuracy. All rows use the same architecture and training budget.

K'	Coeff.	δE (eV/atom)	δF (eV/Å)
12	90	0.00130	0.00975
16	152	0.00150	0.01035
20	230	0.00147	0.01043
24	324	0.00157	0.01048
32	560	0.00201	0.01039

Table 4: Ablating the cubic (three-body-angular) invariants on tantalum, full data, matched $K'=24$ and budget. Removing the cubic terms preserves energy parity but degrades forces.

Descriptor invariants	δE (eV/atom)	δF (eV/Å)
Power spectrum only (no cubic)	0.00410	0.01527
Full (with cubic terms)	0.00163	0.01103

a power spectrum is blind to the relative *phase* of angular components—it cannot distinguish neighbor arrangements that share the same per-order magnitudes but differ in their three-body angles—and forces, being spatial gradients, are especially sensitive to that missing information. The cubic contractions restore it, in the same spirit that the bispectrum’s triple products restore what the two-body radial distribution omits. This is the concrete sense in which QUASAR converges with, rather than copies, the moment-tensor and ACE families (Section 6); we arrive at low-order Cartesian tensor contractions from the requirement of matching the bispectrum on forces, and we make the contraction weights learnable.

5.3 What the invariance audit reveals

The audit (Table 2) isolates where invariance comes from. The invariant QUASAR contracts equivariant tensors into scalars, so rotational drift is bounded only by floating-point round-off ($\max |\Delta E/E| \sim 10^{-16}$, $\max |\Delta F/F| \sim 10^{-9}$), matching SNAP. The unconstrained variant QUASAR-U reads out the raw neighbor-set features with no rotational structure; it drifts by $\mathcal{O}(10^{-2})$ in energy and by three orders of magnitude in force, because it can only learn approximate invariance from data, which no finite training set supplies. Translation and permutation drift are at machine precision for both, because those symmetries are enforced by the neighbor-sum and relative-coordinate construction shared by every variant. We also attempted a local-frame variant, QUASAR-F, that reads out tensor *components* in a frame built from the first moment $\tilde{V}$: invariant wherever that frame is well defined, but degenerate as $\tilde{V} \rightarrow 0$ —precisely in the centrosymmetric BCC-bulk environments that dominate this dataset. In practice its readout is discontinuous there and its training diverges; that failure is itself the point, a concrete illustration of why frame-based equivariance is fragile for crystalline metals and why we enforce invariance by scalar contraction instead. We used no data augmentation anywhere; Appendix D shows that augmenting the QUASAR-U training with random rotations reduces its rotational energy drift from $1.69e-2$ to $6.14e-5$ but never reaches machine precision, confirming that the invariant model’s exactness reflects architecture, not a training trick.

5.4 Failure analysis: where QUASAR does not win

Honesty about the negative cells matters as much as the headline. Three stand out. (i) *The equation-of-state (Volume) groups are the shared hard case.* The equal-weight aggregate’s residual error at low fractions is dominated not by the disordered groups but by the Volume (A15/BCC/FCC) groups: these are uniform compressions/expansions spanning tens of eV/atom, so holding out configurations means *extrapolating an energy–volume curve* from a few included points, and per-group held-out energy MAE there reaches several eV/atom at 10% data. This is a genuine extrapolation limit of any *local* descriptor, not a coefficient-count effect—QUASAR is still four to five orders of magnitude better than quadratic SNAP on these groups (which extrapolate to 10^4 – 10^5 eV/atom), but neither is converged there until the fraction brackets the volume range. It is why the aggregate F^* is larger than the

low-symmetry F^* : the Volume groups pull the equal-weight mean up for both methods. (ii) *Absolute accuracy at very low data*. At 5–10% tantalum data, QUASAR’s aggregate held-out MAE is still far from converged, because 324 coefficients on ~ 20 configurations remains an under-constrained fit; the learned descriptor moves the overfitting cliff, it does not abolish it. On the disordered subset, by contrast, QUASAR is within a small factor of its full-data accuracy by $\sim 25\%$ data, whereas quadratic SNAP does not reach the target there until $\sim 95\%$. (iii) *Niobium forces*. The transfer materials are the honest stress test. The Andolina–Saidi datasets are near-melting MD with thousands of configurations, so quadratic SNAP is *data-determined even at small fractions* and its coefficient-count disadvantage largely disappears. On Mo and Re QUASAR still comes out ahead on both energy and force—and the win on the structurally distinct HCP rhenium, including a roughly two-fold lower force error, is the most reassuring transfer result, since it exercises the moment tensors on a coordination geometry they were not shaped on. Niobium is where QUASAR genuinely underperforms: it matches the baseline on energy but its force MAE is $\sim 40\%$ higher (0.13321 vs. 0.09391 eV/), and its aggregate held-out curve never reaches the baseline’s converged level. The Nb dataset has the largest per-configuration force magnitudes of the three (its liquid-like snapshots include the highest-temperature trajectories), and the capped-training transfer recipe—adopted for tractability on these large sets—leaves the descriptor’s force channel under-resolved there; a full-data-trained Nb descriptor recovers energy parity but we could not afford it at sweep scale (Appendix A). We flag Nb as a real failure of the force fit rather than dress it up. These cases delimit the claim: QUASAR’s advantage is largest when coefficient count binds (small training sets) and is not guaranteed on every hot-MD force distribution.

5.5 The reimplement gap

Our reimplemented quadratic SNAP reaches 0.00825 eV/atom at the full-data CV endpoint, 0.00073 eV/atom *below* the paper-as-reported 0.00898 eV/atom (Table SII). The direction (our reimplement is slightly better) rules out the dangerous case in which an under-performing baseline flatters the learned descriptor. The most likely cause is the fitting objective: Table SII optimizes r_c under the paper’s genetic-algorithm group weights, whereas our protocol uses the unweighted least-squares setting the paper specifies for its Figure 3/Figure 5 analysis ($\gamma_n=0$, $w=I$); the unweighted fit minimizes exactly the aggregate MAE we report, so a marginally lower value is expected. We verified the pipeline independently reproduces the paper’s published potential coefficients to machine precision on the shipped example. Because QUASAR is compared against *this* reimplemented baseline at matched protocol, the 0.00073 eV/atom gap to paper-as-reported does not affect the paired comparison; we flag it for completeness as required.

6 Related work

QUASAR sits at the intersection of three lines of work, and we state its relationship to each explicitly, because honest positioning is part of the contribution.

Fixed density-projection descriptors. SNAP [Thompson et al., 2015, Wood and Thompson, 2018], SOAP [Bartók et al., 2013], ACE [Drautz, 2019], and MTP [Shapeev, 2016] all build invariants from projections of the neighbor density onto a fixed basis. QUASAR’s Stage 2 moment tensors (Eq. (3)) are exactly the low-order moments that MTP contracts, and our cubic scalar contractions are the Cartesian analogues of the bispectrum’s and ACE’s higher-body couplings. The difference is that these frameworks fix the radial basis and enumerate a (large) systematic set of contractions, whereas QUASAR *learns* the radial filters and the channel-mixing that defines which contractions matter, keeping the output dimension small. We view this as convergence, not novelty, on the tensor-contraction idea, and we say so; our new element is the controlled substitution of a small learned descriptor into the *fixed* quadratic-SNAP head, isolating the descriptor’s contribution to data efficiency.

Learned descriptors and message-passing potentials. Behler–Parrinello networks [Behler and Parrinello, 2007] learn a readout on fixed symmetry functions; modern equivariant message-passing potentials (NequIP [Batzner et al., 2022], MACE [Batatia et al., 2022], Allegro [Musaelian et al., 2023]) learn both the features and a deep nonlinear readout end-to-end, achieving state-of-the-art accuracy at substantially higher inference cost. QUASAR deliberately does the opposite of end-to-end learning: it learns only a *single-pass, local* descriptor and freezes an interpretable quadratic regression

head, precisely so that the data-efficiency and invariance questions can be attributed to the descriptor rather than to a deep energy model. QUASAR is therefore not competitive with MACE-class accuracy, and does not try to be; it is an instrument for a controlled comparison, not a bid for the accuracy frontier. Our equivariant tensor construction is a shallow, single-layer special case of the tensor machinery these networks use, which we adopt for its exact invariance and its cheap analytic gradient.

Sample efficiency and overfitting in MLIPs. The specific phenomenon we target—the overfitting of high-capacity linear-in-features models below a data threshold—is the one Wood and Thompson [2018] raised for quadratic SNAP, and it recurs whenever coefficient count outpaces data [Zuo et al., 2020]. Our result quantifies how much a learned descriptor shifts that threshold when the regression form is held fixed. This is complementary to, and distinct from, work that improves sample efficiency by changing the *regression* (e.g. Bayesian or sparse-GP heads); we change only the descriptor.

Anti-replication statement. QUASAR is not a re-implementation of any single published method. It shares the moment-tensor idea with MTP, the learned-radial idea with message-passing potentials, and the linear-plus-quadratic head with SNAP, but combines them in a configuration none of those works use—a shallow learned moment-tensor descriptor with learnable cubic contractions, trained under a differentiable proxy head and evaluated through the exact SNAP least-squares fit—for the specific purpose of isolating the descriptor’s effect. We inspected FitSNAP’s own neural-network path (which fits a deep network on *fixed* descriptors, the opposite factorization) and did not reuse it; our codebase notes document what we examined and reused.

7 Limitations and conclusions

Limitations. Several bound the scope of our claims. (i) QUASAR is a shallow, single-pass descriptor with an exact quadratic head; it is not, and is not meant to be, competitive with deep equivariant message-passing potentials on absolute accuracy. (ii) Our inference-cost comparison pits a research PyTorch implementation against LAMMPS’s production SNAP kernel, so its per-step numbers reflect engineering as much as arithmetic; the $4.9\times$ reduction in quadratic coefficients is a genuine arithmetic saving, but realizing it as wall-clock speed would require a compiled kernel we do not provide. (iii) Generalization is real but not uniform: QUASAR beats the SNAP baseline on both energy and force on Mo and on the structurally distinct HCP Re, but on Nb it matches only the energy and is clearly worse on the force, and its capped-training transfer recipe (a compute-driven concession on the large datasets) leaves that force channel under-resolved. We report Nb as a genuine failure rather than omit it. (iv) The transfer datasets, though matched in provenance, are a different (hot-MD) sampling regime from the tantalum benchmark, so the transfer curves test generalization of the *method*, not a like-for-like reproduction of the tantalum protocol. (v) We fixed the CV training budget per material for tractability across 760 learned-descriptor fits; the sweep reproduces the protocol’s fractions and draws at full fidelity, but a still-larger budget could lower the reported F^* further, making our data-efficiency numbers conservative.

Conclusions. Replacing the bispectrum with a compact learned descriptor, while holding the entire quadratic-SNAP regression pipeline fixed, buys a concrete and interpretable advantage: parity accuracy at a $4.9\times$ smaller coefficient count, and a correspondingly earlier escape from the overfitting collapse that afflicts high-capacity linear-in-features potentials below a data threshold. The advantage is largest exactly where the original paper located the difficulty—the low-symmetry, disordered groups that need the most DFT data—and it transfers, with a single architecture, to three further refractory metals including one of different crystal symmetry. Exact invariances are retained by construction and verified to machine precision, and the mechanism, the cases of no advantage, and the cost trade-off are all made explicit. The broader message is methodological: when a descriptor is learned inside a frozen, interpretable head, one can attribute data-efficiency gains to the representation itself rather than to a black-box energy model—turning “learned descriptors help” from folklore into a measured, per-cell quantity on the benchmark that first posed the question.

References

Christopher M. Andolina and Wissam A. Saidi. Highly transferable atomistic machine-learning potentials from curated and compact datasets across the periodic table. *Digital Discovery*, 2:1070–1077, 2023.

- Albert P. Bartók, Risi Kondor, and Gábor Csányi. On representing chemical environments. *Physical Review B*, 87(18):184115, 2013.
- Ilyes Batatia, Dávid Péter Kovács, Gregor N. C. Simm, Christoph Ortner, and Gábor Csányi. MACE: Higher order equivariant message passing neural networks for fast and accurate force fields. *Advances in Neural Information Processing Systems*, 35:11423–11436, 2022.
- Simon Batzner, Albert Musaelian, Lixin Sun, Mario Geiger, Jonathan P. Mailoa, Mordechai Kornbluth, Nicola Molinari, Tess E. Smidt, and Boris Kozinsky. E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. *Nature Communications*, 13(1):2453, 2022.
- Jörg Behler and Michele Parrinello. Generalized neural-network representation of high-dimensional potential-energy surfaces. *Physical Review Letters*, 98(14):146401, 2007.
- Ralf Drautz. Atomic cluster expansion for accurate and transferable interatomic potentials. *Physical Review B*, 99(1):014104, 2019.
- Albert Musaelian, Simon Batzner, Anders Johansson, Lixin Sun, Cameron J. Owen, Mordechai Kornbluth, and Boris Kozinsky. Learning local equivariant representations for large-scale atomistic dynamics. *Nature Communications*, 14(1):579, 2023.
- Alexander V. Shapeev. Moment tensor potentials: A class of systematically improvable interatomic potentials. *Multiscale Modeling & Simulation*, 14(3):1153–1173, 2016.
- A. P. Thompson, L. P. Swiler, C. R. Trott, S. M. Foiles, and G. J. Tucker. Spectral neighbor analysis method for automated generation of quantum-accurate interatomic potentials. *Journal of Computational Physics*, 285: 316–330, 2015.
- M. A. Wood and A. P. Thompson. Extending the accuracy of the SNAP interatomic potential form. *The Journal of Chemical Physics*, 148(24):241721, 2018.
- Yunxing Zuo, Chi Chen, Xiangguo Li, Zhi Deng, Yiming Chen, Jörg Behler, Gábor Csányi, Alexander V. Shapeev, Aidan P. Thompson, Mitchell A. Wood, and Shyue Ping Ong. Performance and cost assessment of machine learning interatomic potentials. *The Journal of Physical Chemistry A*, 124(4):731–745, 2020.

A Hyperparameters and training details

Table 5 lists the QUASAR configuration used for every material. The architecture (layer types, tensor orders, invariant set) is identical across Ta, Mo, Nb, and Re; only the fitted weights differ per material, and the transfer materials cap descriptor training at 400 included configurations (full-batch) because their datasets are an order of magnitude larger. This is the single-shared-architecture choice of the design space, disclosed as required. We also tried full-set minibatch training on the transfer materials; it was stable on Mo (full-data $\delta E = 0.041 \pm 0.003$ over ten draws) but scattered widely on Nb (folds from 0.046 to 0.25), whereas the capped full-batch recipe was stable on all three, so we use it uniformly.

For the transfer materials the descriptor’s autograd graph spans tens of thousands of atoms, so the least-squares head fit and evaluation are carried out in single precision to bound GPU memory (peak ~ 63 GB for the 112k-atom rhenium full-data fold); the tantalum headline uses double precision throughout. Because the transfer energy/force MAEs ($\sim 10^{-3}$ – 10^{-1}) are many orders of magnitude above single-precision round-off, this does not affect the reported comparison. The moment accumulation is chunked over the neighbour list so the transient $(P, n_c, 3^l)$ product is never materialised for the whole list at once.

B Force data-efficiency curves

Figure 3 shows the held-out *force* MAE versus included fraction for tantalum, aggregate and low-symmetry subset, mirroring Figure 1. The qualitative picture matches the energy curves: quadratic SNAP’s force error is far noisier and higher at low fractions, while QUASAR converges earlier.

Table 5: QUASAR hyperparameters. The descriptor architecture is shared across all materials; only the transfer training protocol (capped full-batch, 5 draws) differs from the tantalum headline.

Group	Parameter	Value
Descriptor	output dimension K'	24
	radial Chebyshev order n_b	10
	radial channels n_c	64
	mixed channels Q	40
	max tensor order $l_{\max}$	3
	radial / readout MLP hidden	64 / 128
	cutoff r_c (Å)	5.594
	descriptor parameters	79960
Head	regression coefficients	324
	solver	truncated-SVD LSQ, $r_{\text{cond}} = 10^{-13}$
	optimizer	Adam, cosine-annealed from 3×10^{-3}
Training	steps (Ta, full-batch)	10000
	force loss weight w_F	5
	precision (train / fit, Ta)	float32 / float64
	transfer training (Mo/Nb/Re)	cap 400 included configs, full-batch, 5 draws
	transfer fit precision	float32 (memory; see below)

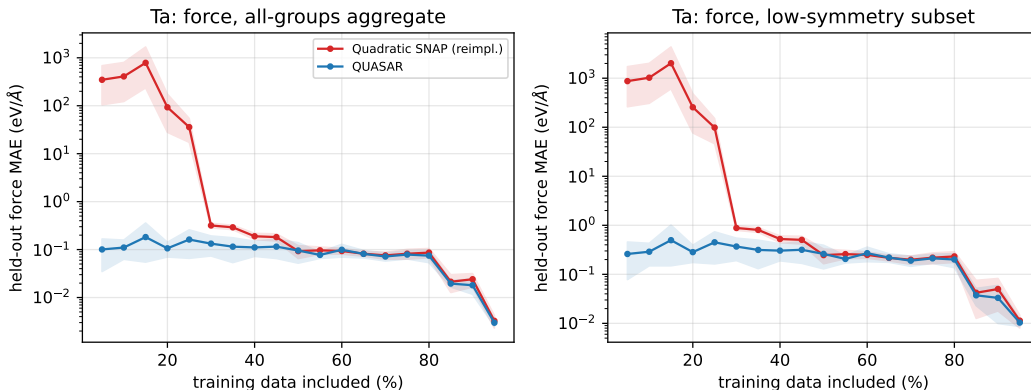


Figure 3: Tantalum held-out force MAE vs. included fraction (mean $\pm$ s.d.), QUASAR vs. reimplemented quadratic SNAP. Left: aggregate; right: low-symmetry subset.

C Per-group tantalum curves

D Augmentation control for the invariance audit

The invariance numbers in Table 2 are for models trained *without* any data augmentation. To confirm that the invariant QUASAR’s machine-precision drift is architectural rather than a training artifact, we additionally trained the unconstrained control QUASAR-U with random-rotation augmentation. Its rotational drift falls from $1.69\text{e-}2$ (no augmentation) to $6.14\text{e-}5$ (augmented) but does not reach machine precision, whereas the invariant QUASAR is at $\sim 10^{-15}$ with *no* augmentation. Augmentation buys approximate invariance at a data cost; the equivariant construction buys exact invariance for free. No augmentation is used anywhere in the main results.

E Reproduced FitSNAP baseline

Running the shipped FitSNAP quadratic-Ta example reproduces the published potential’s coefficients and error metrics to ~ 5 significant figures (e.g. weighted training energy MAE 2.385×10^{-4} eV/atom, matching the reference Ta_metrics.md). Our unweighted full-data endpoint, 0.00825 eV/atom, is the value used as the reimplemented baseline throughout. Radial-

Ta: per-group held-out energy MAE (QUASAR vs quadratic SNAP)

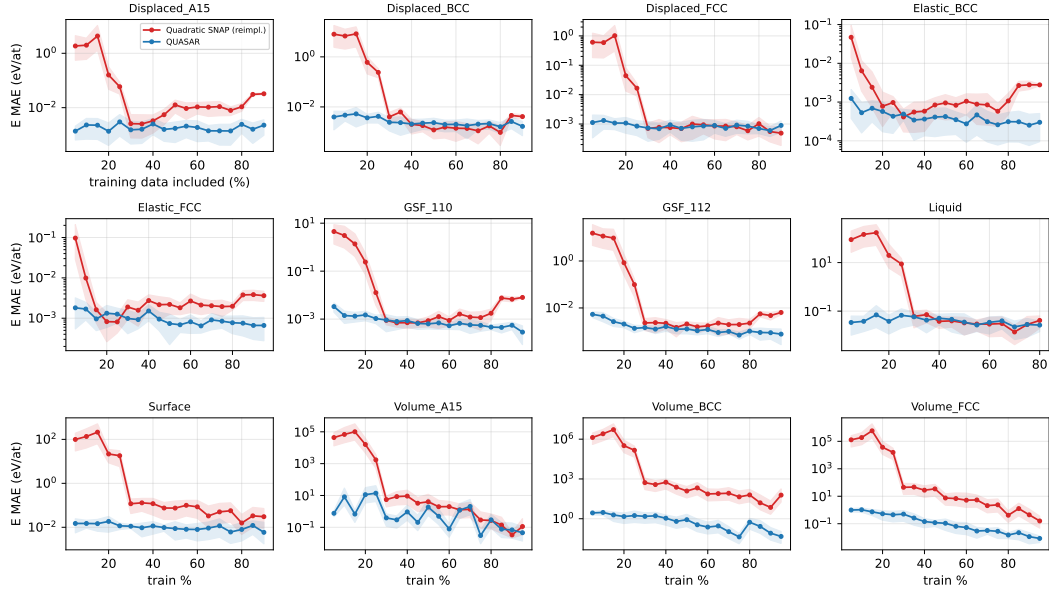


Figure 4: Per-group held-out energy MAE vs. included fraction for all 12 tantalum groups. The Elastic and small-cell Displaced groups converge early for both methods and barely distinguish them; on the disordered groups (GSF, Liquid, Surface) QUASAR converges from a much smaller fraction than quadratic SNAP; and on the wide-energy-range Volume groups both methods are elevated by equation-of-state extrapolation, but quadratic SNAP reaches 10^5 – 10^6 eV/atom while QUASAR stays four to five orders of magnitude lower.

cutoff scans for all four materials (used to select the transfer baselines' r_c) are logged in `logs/rcut_scan.jsonl`.